

Linking Membrane Organisation to Light-Harvesting Regulation in LHCII Membranes

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ABSTRACT

Photosynthetic light harvesting is governed by collective excitonic interactions within densely packed pigment–protein membranes, where nanoscale organisation shapes both energy transfer and photoprotective dissipation. In plants, the major antenna complex LHCII sits at the centre of this balance, yet the causal link between lateral membrane organisation and excitation-energy dynamics remains poorly resolved. Here I present two complementary strategies to interrogate LHCII organisation in model photosynthetic membranes and how it relates it to energy transfer and dissipation.

First, in-membrane electrophoresis (IME) applies an in-plane electric field across a supported lipid bilayer (SLB), redistributing charged membrane components to generate continuous concentration gradients within a single membrane [Meredith et al. 2023]. By using IME to reorganise LHCII in polymer confined patterned SLBs, Fluorescence Lifetime Imaging Microscopy (FLIM) reveals that local increases in protein density shorten the fluorescence lifetime, consistent with the onset of energy-dissipative quenching. This isolates lateral organisation as a control parameter, showing that antenna density alone is sufficient to tune LHCII between light-harvesting and dissipative states within a single, continuously graded membrane.

Second, I will present the method of Structured Excitation Energy Transfer (StrEET) microscopy to quantify excitation transport in LHCII-containing SLBs. StrEET follows the decay of a spatially patterned excitation, giving direct access to exciton transport. Distributing excitation over a large area lowers the fluence by up to four orders of magnitude relative to point-excitation spatiotemporal microscopy, suppressing singlet–singlet annihilation and photodamage — critical for fragile light-harvesting protein systems [Brinatti et al. 2024]. Applied to LHCII bilayers, StrEET provides a direct, low-perturbation measure of the exciton diffusion length, allowing energy transport to be correlated with the local protein density and quenching resolved by FLIM.

REFERENCES

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